SORAIA PEDRO TARRINHA

Software Engineer | Python Data Engineering | SQL Analytics | Data Quality & ETL

+351 938 790 910 soraiatarrinha@gmail.com https://linkedin.com/in/soraia-tarrinha-4b701813
https://github.com/soraiatarrinha97 Tomar, Portugal

SUMMARY

Software engineer and data professional with 6+ years of experience in Python, advanced SQL/PLSQL and Oracle, developing data ingestion, transformation, validation and reporting workflows. Skilled in SQL performance optimization, automation, data quality, reconciliation and ad-hoc analysis. Combines hands-on technical expertise with cross-functional collaboration, stakeholder engagement and additional Release Manager responsibilities.

EXPERIENCE

Software Engineer – Python, SQL/PLSQL & Data Analytics

SOFTINSA (IBM subsidiary) 02/2020 - Present Remote

Vodafone / Orange Spain / Central de Cervejas (Grupo SCC)

- Developed and maintained daily and monthly data integration workflows for Vodafone's commission project, using Python, SQL/PLSQL and Oracle to process CSV files containing 15,000+ rows per execution, apply business rules, validate and transform data, and load staging and target tables.
- Analyzed and troubleshot complex datasets with Oracle SQL/PL/SQL, developing and maintaining stored procedures, packages, and functions with CTEs, recursive queries, ROW_NUMBER, and MERGE for incremental and large-scale transformations. Documented modification history in scripts and procedures, including author, change details, and date.
- Wrote and supported modular Python and Shell scripts on Linux for data processing, automation, CSV integration, and recurring operational workflows; used separate functions, logging, and inline documentation to improve maintainability, while reducing repeated database queries by retrieving data once and reusing it in memory.
- Resolved more than 30 production defects in 2025 involving commission calculations, saved-import queries, missing calculation elements, LEFT joins, and restrictions; performed root-cause analysis and supplied evidence for confirmation.
- Enhanced and supported IBM Varicent ICM commission reports using SQL and Oracle, working with data sources, partitions, restrictions, formulas, and aggregations. Analyzed business financials for Vodafone Finance, investigated discrepancies, and delivered ad-hoc reporting to validate calculated commission values.
- Collaborated with Vodafone Finance and technical teams on commission discrepancies, client approvals, data validation, testing, UAT, release activities, and data governance. Documented processes in Confluence, recorded evidence in Excel and Jira, compared actual and expected results, and tracked releases; the AMO team performed production installation.
- Partnered with a 6-person Scrum team, a 4-person Agile team, and a 4-person AMO team to analyze SQL data for actionable insights on commission processes, while guiding 3 junior team members in SQL.
- Supported KPI management by configuring and maintaining mappings between KPIs and commission rules in Varicent ICM, according to Vodafone's business requirements.

Release Manager (Additional Responsibility)

SOFTINSA (IBM subsidiary) 03/2021 - Present Remote

- Conducted release-readiness reviews, verifying Jira evidence, test results, technical dependencies and client approvals before including changes in release packages.
- Coordinated biweekly data release packages during the Release Manager rotation, overseeing delivery activities across a 6-person Scrum team, a 4-person Agile team, and a 4-person AMO team responsible for installation.
- Validated Python and SQL changes in testing and pre-production environments and designed verification checks before handoff, contributing to CI/CD-related delivery workflows. Coordinated with development, functional, and deployment teams on production deliveries; the authorized AMO team performed production deployment.
- Managed Git branches and merges, deployment procedures, and version traceability. Organized structured testing evidence by documenting validation results and attaching supporting evidence to Jira issues; tracked planned changes, risks and release readiness.
- Managed releases monthly with engineering duties: coordinated release readiness, risk identification, documentation, validation.

KEY ACHIEVEMENTS



Reduced execution time of a critical Oracle process by about 60%

Cutting runtime from 30 to 12 minutes through SQL redesign and performance optimization, enabling calculated values to reach the Finance team faster.



Python Automation Implementation

Implemented Python scripts that streamlined data processing, reducing manual efforts by over 30 hours monthly.

SKILLS

Reporting & Data Quality

Data Quality, Data Validation, Data Reconciliation, Root Cause Analysis, Data Quality Tools, Data Visualization, Data Reporting, Report Development, Query-Based Reporting, Ad-hoc Reporting

Programming & Databases

Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, Shell Scripting, Bash, Oracle, SQL, SQL Server (MSSQL), PostgreSQL, PL/SQL, Stored Procedures, Packages, Functions, CTEs, Recursive Queries, ROW_NUMBER, MERGE, Query Optimization, Data Frames, Software design principles

Data Engineering & Analytics

ETL, Batch Data Pipelines, Data Ingestion, Data Transformation, Staging Workflows, Incremental Loads, API Integration, JSON/CSV Processing, Business Rule Validation, Data Modeling, Data cleaning, Data Analysis, Exploratory Data Analysis, K-Means clustering, PCA, REST APIs, BeautifulSoup, Requests

Development Tools & Collaboration

UAT, Integration Testing, Code Review, Confluence, Test Case Design, Git, Version control, Jira, Agile/Scrum, Release Coordination, Pre-production Validation, Production Support, Technical Documentation, Cross-functional Collaboration, Business Rule Translation, Bitbucket, IBM Varicent Incentive Compensation Management (ICM), Linux, CI/CD (GitHub Actions CI; release and deployment coordination)

TRAINING / COURSES

AWS Certified Solutions Architect Associate (SAA-C03) - In Progress

Udemy

LANGUAGES

Portuguese	Native
English	C1
Spanish	B2

EXPERIENCE

AI / Machine Learning Developer Intern - SADGE

PRIMELAYER | 02/2019 - 06/2019 | Coimbra, Portugal

Final-year Bachelor's internship completed in Coimbra as part of the B.Sc. in Computer Engineering - Information Systems

- Engineered a Python-driven forecasting module for Primelayer's SADGE platform, transforming 756,908 IPMA time-series observations and river-flow data into actionable insights for flood-risk management.
- Evaluated LSTM and regression models in TensorFlow/Keras and Scikit-learn, applying preprocessing, normalization, train/test splits, parameter tuning, data profiling, and RMSE assessment.
- Unified time-series sensor, hydrometric, and meteorological datasets with Pandas, NumPy, and PostgreSQL, transforming and persisting prediction data for downstream platform use.
- Produced a model adopted on Primelayer infrastructure to generate next-day river-flow forecasts, moving the work beyond experimentation into technical delivery.

EDUCATION

MSc in Business Intelligence and Analytics

Polytechnic University of Tomar | 2021 - 2024 | GPA **16** / 20

- Selected projects using Jupyter Notebooks: automated Python data ingestion from external weather and social-media APIs with JSON processing; NLP and sentiment analysis; data preprocessing and analysis with Pandas/NumPy; PCA, K-Means, Logistic Regression and SVM

B.Sc. in Computer Engineering - Information Systems

Polytechnic University of Coimbra - ISEC | 2015 - 2019

- Data Integration, Database Architecture and Administration, Information Systems and Artificial Intelligence

PUBLICATIONS

Gamification in the Context of Cultural Heritage in Portugal: Analysing the Perspectives of Tour Operators and Tourists Using Python

WorldCIST 2025 conference paper in Emerging Trends in Information Systems and Technologies (Springer), pp. 151-162

Python-based analysis and visualization of tourist and tour-operator survey data. DOI: 10.1007/978-3-032-01130-5_13

PROJECTS

Weather Data Pipeline & Exploratory Analysis

12/2021 - 01/2022

GitHub: https://github.com/soraiatarrinha97/weather-data-pipeline

Python, pandas DataFrames, REST APIs, ETL, Folium, Matplotlib, pytest, GitHub Actions

Originally developed as an academic project in 2021–2022 and modernized in 2026: built a Python ETL pipeline integrating the OpenWeather REST API to extract, validate, transform and store weather observations, processing data with pandas DataFrames.

Performed exploratory analysis of 1,243 historical observations, adding Folium and Matplotlib visualizations, pytest automated tests and GitHub Actions CI. This demonstrates the CI side of CI/CD; the project does not claim automatic continuous deployment.